

MA 301

Test 3

Fall - 2013

Problem	points	out of
1		10
2		10
3		12
Score		32

Name: _____

HR: _____

Signature: _____

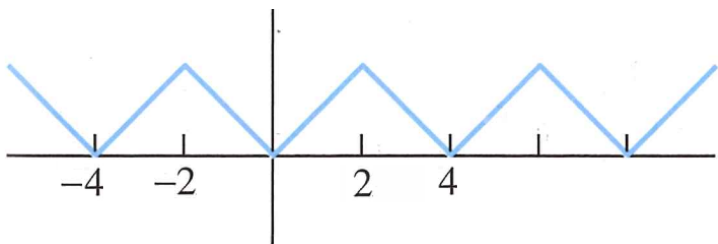
Formula Sheet

Fourier Series: $f(x) = \sum_{n=-\infty}^{\infty} d_n e^{i\pi n x / L}$ where $d_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-i\pi n x / L} dx$

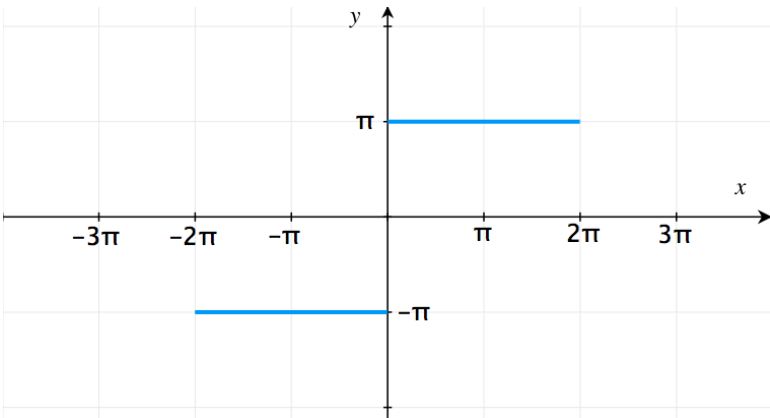
if $f(x)$ is even then $d_n = \frac{1}{L} \int_0^L f(x) \cos\left(\frac{\pi n x}{L}\right) dx$

if $f(x)$ is odd then $d_n = \frac{-i}{L} \int_0^L f(x) \sin\left(\frac{\pi n x}{L}\right) dx$

Problem 1: Determine the Fourier Series of the sawtooth function below.



Problem 2: Determine the Fourier Series of the function depicted below.
Assume it periodically extends along the x -axis.

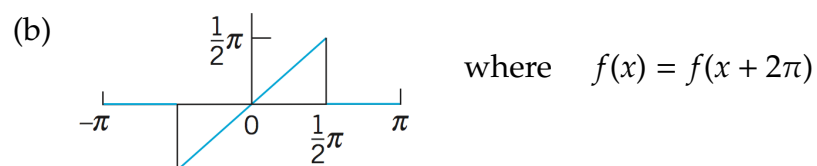


Problem 3: For each function $f(x)$ below with specified period, Setup (DO NOT Compute) the integral (or sum of integrals) needed to compute the Fourier coefficients. $f(x)$ should not be in your final answer.

(a) $f(x) = \begin{cases} 0 & -2 \leq x \leq 0 \\ 1 & 0 \leq x \leq 1 \\ 0 & 1 \leq x \leq 2 \end{cases}$ where $f(x) = f(x + 4)$

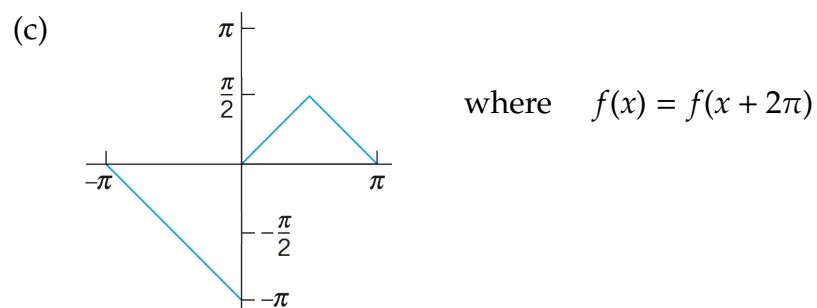
$$d_0 =$$

$$d_n =$$



$$d_0 =$$

$$d_n =$$



$$d_0 =$$

$$d_n =$$